

Assignment #1. HTML & CSS Basics

Objective:

By the end of this assignment, students will be able to:

- build a small, multi-page website using well-structured, semantic HTML5;
- integrate forms, a data table and multimedia into their pages correctly;
- make their pages accessible and standards-compliant, and prove it with the W3C validator;
- apply CSS fundamentals (selectors, colors, fonts, spacing, alignment, classes & IDs) from an external stylesheet;
- organize a project into folders, publish it with GitHub Pages, and defend their own code.

Rules for Submission:

1. This is an individual assignment. **If the works of different students are the same, the grade is 0.**
2. Submit before the deadline. **Late works are not accepted!** To get a grade you must **submit before the deadline AND defend your work at the practice lesson.**
3. You must submit: your report in **Assignment1_YourName.docx** (or .pdf) format, and your project FOLDER zipped with all needed files in **Assignment1_YourName.zip** format. **If you submit in other formats, your grade will be 0.**
4. **IN THE REPORT MUST BE:**
 - All parts with a short description of what you did step-by-step
 - Screenshots of your pages and code
 - A short conclusion about what you learned
 - The **live GitHub Pages URL** of the published web site

Environment Preparation:

1. Install VS Code (or another code editor): <https://code.visualstudio.com/download>
2. Create a folder for the project (e.g., assignment1)

Part 1. Project setup and home page

Choose **any topic you are interested in** and decide what information you would like to present to the visitor.

Task 1. Create a basic webpage on a topic of your choice. (10 points)

- Create the folder structure: **index.html**, **about.html**, **contact.html**, **css/style.css**, an **images** folder and a **media** folder.
- Write a valid HTML5 boilerplate in every page (with lang, charset and the viewport meta). Create **css/style.css** but leave it empty for now (you will fill it in Part 3).
- Divide **index.html** into header, nav, main and footer.
- Inside main put one h1, two h2 sections with your own text, one unordered and one ordered list, and two images (from the images folder) with meaningful alt text.

Output expectation: pages open in the browser with no errors in the console.

Part 2. Advanced HTML

Task 2. About page, media and navigation (20 points)

Build **about.html** about one thing from your topic: a person, a place, a product or a track.

- A heading and a paragraph of 3-5 sentences about your subject.
- An ordered list of 4-6 key dates and an unordered list of at least 4 facts.
- A figure with a photo and a figcaption (descriptive alt text on the image).
- A blockquote with the source written under it, and two or three words marked with strong or em.
- One embedded multimedia element that fits your topic: audio, video or an iframe with a short label.
- An external link that opens in a new tab (target="_blank" rel="noopener").

Navigation bar. Make one navbar with links to all your pages.

- Put the same navbar in the header of every page and check that every link works.
- Mark the current page (for example an active class).

Task 3. Data table and contact form (20 points)

Data table. Make a table with real data about your topic: a price list, a timetable, a comparison, a schedule or a discography.

- At least 4 columns and 5 rows of real data.
- Use caption, thead, tbody and tfoot.
- Use th for header cells and add scope to each of them.
- Join at least two cells with colspan or rowspan where it makes sense.

Contact form. Build a **functional HTML form** in **contact.html** (no backend is needed, the form does not have to send data).

- Wrap the fields in a fieldset with a legend.
- Every field has a label connected to its input with matching for and id.
- At least five different input types (for example text, email, number, date, color, checkbox, radio, textarea) plus a submit button.
- Use required on at least two fields and one more native validation attribute (placeholder, min/max, maxlength or pattern).
- Group related radio buttons with the same name.

Part 3. CSS styling

Task 4. CSS styling (10 points)

Link **css/style.css** to all pages and style them from this single external stylesheet. No style blocks and no inline style attributes.

- Choose 3-4 colors and write them in at least three formats (name, hex, rgb or hsl).
- Choose fonts and set sizes, weights and alignment (maximum two font families).
- Use margin, padding and border so you can see the box model in the developer tools.
- Use element, class and ID selectors. Use the ID only for one unique element.
- Style the navigation (no underline, good spacing, a hover effect that changes more than the color), the table (readable rows) and the form (a visible **:focus** state).
- Use at least three CSS units (px, %, rem, em, vh) and explain the purpose of each of them in your report.

Part 4. Publishing your website

Task 5. Validate and publish (5 points)

- Validate every HTML page with the W3C Markup Validator (https://validator.w3.org/#validate_by_input) and the stylesheet with the W3C CSS Validator (https://jigsaw.w3.org/css-validator/#validate_by_input); fix the errors and put the clean-result screenshots in your report.
- Publish the site on GitHub Pages with **index.html** as the entry page.
- Put the live link in your report and check that every page and every navbar link works online.

Assessment criteria

Criterion	What is checked	Points
Task 1. Project setup and home page	- correct folder structure and valid HTML5 boilerplate in every page - header, nav, main and footer used correctly, with one h1, two h2 sections, both lists and two images with meaningful alt text - pages open in the browser without errors in the console	10
Task 2. About page, media and navigation	- heading and 3–5 sentence paragraph, ordered and unordered lists - figure with figcaption and descriptive alt text, blockquote with source, and meaningful use of strong or em - working multimedia element with an appropriate label - external link opens in a new tab with target="_blank" and rel="noopener" - same navbar is used on every page, all links work, and the current page is marked	20
Task 3. Data table and contact form	- table contains real data with caption, thead, tbody and tfoot, at least 4 columns and 5 rows - th cells have scope, and colspan or rowspan is used appropriately - form uses fieldset and legend, labels are correctly connected to fields, and at least five different input types are used - at least two fields use required, plus one additional native validation attribute - related radio buttons are grouped using the same name	20
Task 4. CSS styling	- all pages use one external stylesheet with no inline styles or style blocks - 3–4 colors are used in at least three formats; fonts, sizes, weights and alignment are applied - margin, padding and border demonstrate the box model - element, class and ID selectors are used correctly; ID is used for one unique element - navigation, table and form are styled, including a visible :focus state and a hover effect that changes more than color - at least three CSS units are used and explained in the report	10
Task 5. Validate and publish	- all HTML pages and the stylesheet pass the appropriate W3C validators; clean-result screenshots are included in the report - website is published on GitHub Pages with index.html as the entry page - live link works and all pages and navbar links work online	5

Defense	- explains the work the student has done and demonstrates the website working live without errors - answers the instructor's questions about their own code and the topic - makes one small change requested by the instructor without assistance	30
Report	- includes name and group, all tasks in order, screenshots and short explanations in the student's own words - includes a short conclusion about what the student learned	5
Total		100

Resources:

1. Abitova G.A. Web technologies Front-End Development. Part 1 (2022).
2. <https://www.w3schools.com/html/>
3. https://www.w3schools.com/html/html_tables.asp
4. https://www.w3schools.com/css/css_selectors.asp
5. <https://learn.astanait.edu.kz>

***Questions will be based on the provided resources**

Note: If you have questions, feel free to come to my office hour in room C1.3.354 (time and date are in Moodle), or write to me by email or in MTeams: [**a.fedenko@astanait.edu.kz**](mailto:a.fedenko@astanait.edu.kz).

And do not be afraid to use the internet: documentation, tutorials and examples help a lot.

Good luck!